Physico-chemical studies on raw and processed moth caterpillar silks from Northern Western Ghats, the mega-biodiversity hotspot of India

Dr. Ravindra D. Chaudhari

Associate Professor & Head, Zoology Research Centre, & In Charge,
P. G. Department of Zoology, Shri Shiv Chhatrapati College, Junnar, Pune,
Affiliated to Savitribai Phule Pune University, Pune

Abstract

The earliest evidence of the lustrous fabric can be traced back to more than 5000 years ago in China, where the fabric was reserved for royalty. Due to sericulture technologies, today even the middle class wear it on occasion. In India, Mulberry silk, produced by *Bombyx mori* moths reared in captivity and Tasar silk, produced by wild *Antheraea* species of moths, is common. The quality of silk fibers produced by different kinds of moths varies considerably from one another. Knowledge of their properties will aid in selecting the most suitable variety for other possible uses. Silk is essentially a protein; it is extremely biocompatible and can be used as sutures, dressings or even tissue implants. The last few decades have seen a surge in non-traditional applications of silk fibers. Silkworm fiber has been identified as a suitable material for biomedical and electronics applications because of its superior optical, mechanical and biological properties.

Herein, I present comparative studies pertaining to the structural and morphological features of naturally harvested moth caterpillar silk fiber samples obtained from domesticated (*Bombyx mori*) as well as wild species, *viz*., *Antheraea mylitta* and *Antheraea papiha*. It has been observed that silk fibers obtained from silk cocoons are several microns in thickness. Surprisingly, wild variety i.e. tasar silk samples show better structural and morphological properties. These fibers may find broad-spectrum applications in biomedical and electronics research.

This study is particularly significant as it throws light on the properties of these silk varieties before and after degumming. Based on the insights gained, experiments for using degummed silk fibers as well as choice of type of silk fibre (wild or domestic) can be carried out keeping in mind the particular application domain.

Biography

Dr. Ravindra D. Chaudhari is working as Associate Prof. and Head, Zoology Research Centre, & In - Charge, P. G. Department of Zoology, Shri Shiv Chhatrapati College of Arts, Commerce

& Science, Junnar, Pune, affiliated to Savitribai Phule Pune University, Pune, Maharashtra, India. He has obtained M. Sc. degree from University of Pune in 1991 and Ph. D. degree in Zoology in 1995. He has working expertise in the field of Entomology, Biodiversity, Microbial diversity, and Wild life for 25 years. He is the Expert Wildlife Biologist & Ecologist member for Eco-sensitive Zone and Critical wildlife habitat committees of Govt. of Maharashtra for five protected reserved forests in Maharashtra State. He has published 40 papers in various reputed peer reviewed international journals. He is the recipient of "**BEST TEACHER AWARD**" (All subjects) given **by University of Pune** for the year 2014-15. Presently he is the Member of "**Board of Studies in Zoology**" of Savitribai Phule Pune University, Pune. Recently, he has started the research work on Nano-biotechnology, biological synthesis of nanomaterials as well as their antimicrobial and other applications. Dr. Chaudhari is having MOU's & research collaboration with 5 foreign universities & few Indian institutes. Dr. Chaudhari in collaboration with Korean scientists has recently published a research paper in the prestigious journal **"Nature publishing groups, Scientific Reports"**.

Photo :



Email id : rdchaudhari2004@yahoo.co.in